

Data-Driven Collision Detection Between Surgical Tools in Pattern-Cutting Training Video

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Abstract—Detecting collisions between instruments is an important step toward automated skill assessment in robotic and simulated surgery. In this work we present a data-driven pipeline that detects, at the level of a single video frame, whether two surgical tool tips are in collision. We use a pre-trained YOLO11n segmentation model to detect the left tip (TIPL), right tip (TIPR) and a tip-with-marker (TIPandCircle) on every frame of a pattern-cutting training video, and turn those detections into a compact set of geometric and temporal features: positions, pairwise distances, bounding-box overlap, segmentation-mask area as a depth cue, and short-window motion. On top of these features we train an XGBoost classifier and evaluate it with 5-fold stratified group cross-validation so no information leaks between neighbouring frames. The system reaches ROC-AUC 0.812 on all 3,917 pooled out-of-fold frames (mean 0.840 ± 0.055 across folds) and recovers 68.5% of true collision frames at 43.0% precision at a recall-favouring operating point. We also describe the main practical limits — a single video, class imbalance (14.2% positive frames), and severe mutual occlusion of the tips at the moment of contact — and outline the changes that would most improve future performance.

Index Terms—surgical skill assessment, collision detection, instance segmentation, YOLO, XGBoost, class imbalance.

- GitHub Repository: https://github.com/ashumishra72/collision-detection_project_CV

I. Introduction

Automated assessment of surgical skill is an active area of research, both for training feedback and for objective evaluation of robotic-assisted procedures. A clear low-level signal of poor technique is a collision between two instruments. Such events are short, visually subtle and often hidden behind the tools themselves, yet supervisors flag them as markers of unsafe or clumsy handling.

Our task, defined by the assessment brief, is: given a surgical training video and a pre-trained tool detection / segmentation model, decide for every frame whether the two tools are in collision. A human-annotated ground truth is also provided, so this is a supervised classification problem. Two things make it hard in practice. First, the provided detector was trained for tool segmentation, not collision, and tends to lose one or both tips at the exact moment of contact — in our data both tips are visible together on only 43.5% of labeled collision frames vs. 79.4% elsewhere. Second, classes are heavily imbalanced (14.2% positive), so a trivial “always negative” classifier already reaches $\approx 85.8\%$ accuracy.

We chose Option 1 of the brief (data-driven). Our contributions are: (i) a compact set of geometric, depth and temporal features built on top of the detector’s outputs, including a constant-velocity tracker that keeps the tip–tip distance meaningful through brief occlusions; (ii) a leakage-safe evaluation using stratified group k -fold that groups neighbouring frames together while balancing the class ratio across folds; and (iii) a global operating point tuned once from the pooled probabilities using an F_β objective with $\beta = 1.3$, which slightly favours recall — a reasonable default for a safety-oriented monitor.

Related work. Deep instance segmentation (Mask R-CNN, YOLO) is the standard for tool detection in surgical video [1], [3], [4]. Skill assessment approaches score whole videos [5] using rating scales such as OSATS [6]; our per-frame collision detector sits upstream of these scorers. Handling class imbalance via cost-sensitive learning [7] and gradient boosting [2] is standard practice for rare-event detection.

II. Materials and Method

A. Pipeline Overview

Fig. 1 shows the full pipeline in six stages: raw inputs (video, segmentation masks, mask-area depth cue, ground truth) $\rightarrow$ per-frame feature extraction (position, distance, depth, motion) $\rightarrow$ XGBoost training with a class-weight for imbalance $\rightarrow$ probability smoothing and a global decision threshold $\rightarrow$ precision/recall/ F_1 evaluation $\rightarrow$ output artefacts (per-frame CSV, charts and paper tables). The following subsections describe each stage in turn.

B. Video, Ground Truth and Detector

The dataset is a single pattern-cutting training video of ≈ 12 min 48 s showing two robotic arms (“arm 2” and “arm 4”) manipulating a cutting pattern with a base board and push pins. The ground truth is a Word-document table with 89 events (64 collision, 7 near-miss, 18 “no true contact”); collisions are mostly swiping (36) or contact / sustained contact (26), plus a few scraping and generic collision events. Timestamps in the document were reformatted by Word into an “M.SS” style (3.09 instead of 3:09); we parse the table directly from the document’s XML and convert to seconds. The tool detector (video_tip_circle_yolo11n_seg_best.pt) is

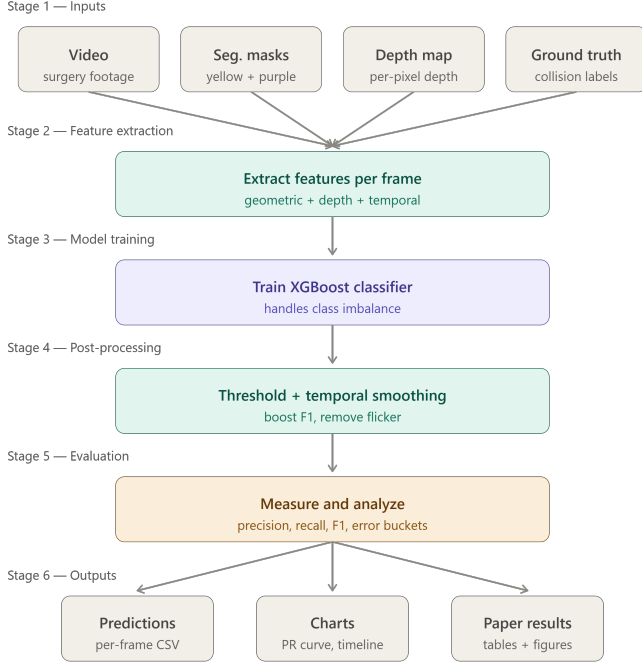


Fig. 1. End-to-end collision-detection pipeline.

a YOLO11n segmentation model with three classes: TIPL, TIPR and TIPandCircle.

Fig. 2 shows a representative frame with the detector’s outputs overlaid: the yellow bounding box marks TIPL (confidence 0.81), the purple box marks TIPR (confidence 0.94), and the green box is TIPandCircle. The top-left overlay reports the frame index, timestamp, and two of the derived features (normalised tip–tip distance and a relative depth difference cue).

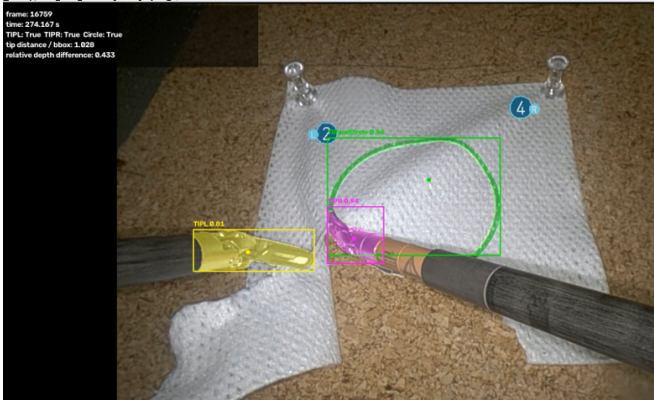


Fig. 2. A sample sampled frame showing the YOLO11n-seg detections (yellow = TIPL, purple = TIPR, green = TIPandCircle). The top-left overlay shows two derived per-frame features used by the classifier.

C. Features

The video is sampled at 5 fps, giving 3,917 frames; 558 (14.25%) fall inside a labeled collision event (with a ± 0.5 s pad to absorb annotation imprecision). For each frame we keep the highest-confidence detection per class and compute 44 features grouped as:

- **Appearance:** presence flag, detection confidence, centroid (x, y) , bbox size (w, h) and area, per class.
- **Pairwise geometry:** centroid distance dist_TIPL_TIPR (main collision proxy), tip-to-marker distances, bbox IoU and mask IoU / overlap fraction between the two tips.
- **Depth cue:** segmentation-mask area of each detection (a monocular pseudo-depth signal — objects appear larger as they approach the camera).
- **Temporal:** frame-to-frame speed of each tip and their sum, delta of the tip–tip distance (closing speed), a smoothed tip–tip distance, a 1-second rolling minimum of that distance, and 1-second rolling maxima of the two overlap measures.
- **Occlusion-robust tracking:** a constant-velocity tracker on each of TIPL and TIPR extrapolates a position for a short time budget when the tip is undetected, yielding $\text{dist_TIPL_TIPR_tracked}$, a track-state indicator, and the time since both tips were simultaneously present.

D. Classifier, Cross-Validation and Threshold

We train an XGBoost binary classifier (binary:logistic, eval metric aucpr) with 800 boosting rounds and early stopping after 40 rounds, tree depth 5, learning rate 0.05, subsample and column subsample 0.9, minimum child weight 2. Class imbalance is handled by setting scale_pos_weight to the training-set negative/positive ratio (≈ 5.6).

We evaluate with Stratified Group k -Fold ($k = 5$): the timeline is first cut into 5-second chunks that stay whole in a single fold (preventing frame-level leakage between train and test), and folds are then assigned so that each fold’s collision rate matches the overall $\approx 14\%$ as closely as possible. After cross-validation we have an out-of-fold probability for every frame. We (i) smooth these probabilities with a centred rolling mean of width 3 to suppress single-frame spikes, then (ii) pick a single global decision threshold by maximising F_β with $\beta = 1.3$ (which weights recall slightly more than precision). The resulting threshold is 0.498.

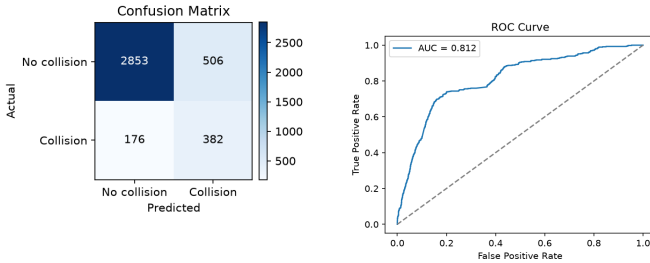
III. Results

Overall. Table I reports the pooled out-of-fold metrics on all 3,917 frames. The system reaches ROC-AUC 0.812 and average precision 0.442; at the chosen operating point it recovers 68.5% of true collision frames at 43.0% precision for $F_1 = 0.528$. Fig. III shows the confusion matrix: 382/558 (68.5%) true collisions correctly flagged, 2,853/3,359 (84.9%) true non-collisions correctly ignored,

and a false-positive rate of 15.1% on negatives. Fig. III shows the ROC curve, which rises steeply for the first portion of the false-positive-rate axis.

TABLE I
Pooled out-of-fold metrics on all 3,917 frames (558 positives).

Metric	Value
Accuracy	0.826
Precision	0.430
Recall	0.685
F ₁ -score	0.528
ROC-AUC	0.812
Average Precision (AP)	0.442
Decision threshold	0.498



Confusion matrix.

ROC curve.

Fig. 3. Evaluation curves at the chosen operating point (threshold = 0.498).

Cross-validation stability. Table II shows the per-fold breakdown. All folds carry 107–121 positives (a similar $\approx 14\%$ rate), confirming that group stratification kept them balanced. Fold 4 is visibly the hardest and fold 3 the easiest — a useful reminder that any single “lucky” chronological split would over- or under-state performance.

TABLE II
Per-fold cross-validation (Stratified Group KFold, 5-second groups).

Fold	n	Pos.	Prec.	Rec.	AUC
1	789	107	0.404	0.533	0.834
2	789	113	0.426	0.761	0.839
3	784	110	0.571	0.845	0.914
4	765	121	0.282	0.504	0.746
5	790	107	0.512	0.794	0.868
Mean	—	—	0.439	0.688	0.840
Std	—	—	± 0.099	± 0.141	± 0.055

Feature importance. Fig. 4 shows the top 15 importances of the final XGBoost model. The raw tip–tip distance `dist_TIPL_TIPR` (0.139) and the tip-and-marker detection confidence `TIPandCircle_conf` (0.115) dominate. The 1-second rolling minimum of the tip–tip distance is third — a sign that the classifier is using short-window temporal context. Mask-area features (e.g. `TIPL_area`) appear in the top five, supporting the intuition that

mask area works as a rough depth cue. The occlusion-robust tracked distance (`dist_TIPL_TIPR_tracked`) also contributes, as intended.

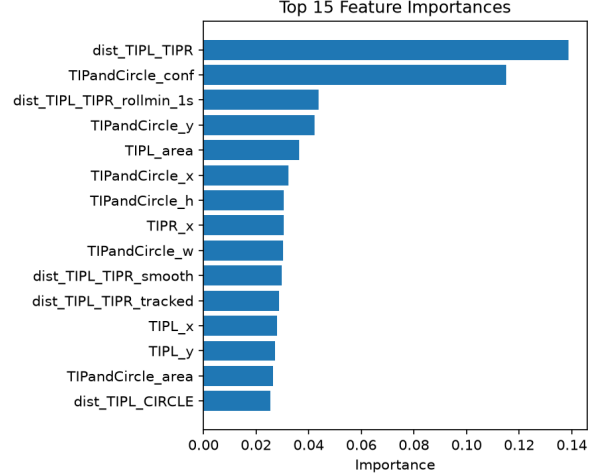


Fig. 4. Top 15 feature importances of the final XGBoost model.

IV. Discussion, Conclusion and Future Work

An ROC-AUC of 0.812 pooled (mean 0.840 ± 0.055 over 5 folds) shows the classifier is a useful ranker on this video. The operating-point precision looks modest, but three things make the problem harder than the raw number suggests. First, the detector loses one or both tips on 56.5% of collision frames because they occlude each other at contact, and every downstream feature inherits this. Second, “collision” as annotated is heterogeneous (swipes, sustained contact, scrapes, push-pin contacts) so a single-head classifier trades off across all of them. Third, many false positives are frames where the tips are close and moving together but the annotator labelled a near-miss — a labelling ambiguity as much as a model failure. False negatives are dominated by deep-occlusion windows in which even the tracked distance has expired.

We view this as a solid baseline for the data-driven route (Option 1 of the brief); its limitations map cleanly to next steps: (i) more videos across sessions and operators, to test cross-video generalisation; (ii) fine-tune the detector with a collision-aware objective (Option 2 of the brief), e.g. a “pair-in-contact” class; (iii) replace the mask-area depth cue with a real monocular or stereo depth estimate; (iv) a lightweight temporal model over the per-frame probabilities for event-level detections; (v) predict collision type instead of a single collapsed label. The largest gains available now come from investing in the detector and in more data — not from further tuning of the classifier alone.

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